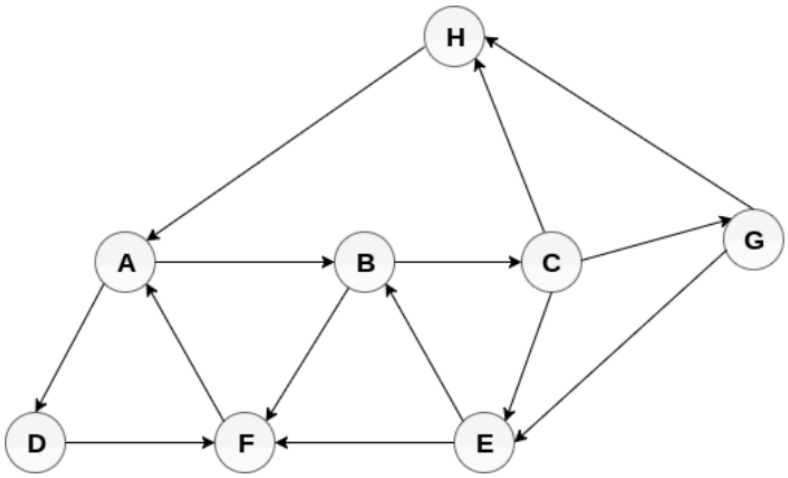
LAB 2

For the given graph 1 and 2 illustrate BFS and DFS search, . Upload the code along with the output.

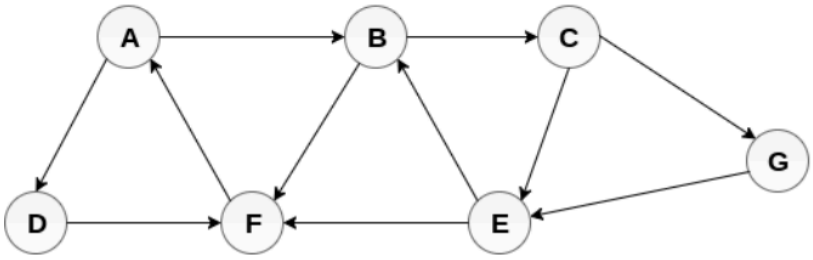
GRAPH1[Start node= H, Goal node= F]



Adjacency Lists

A : B, D
B : C, F
C : E, G, H
G : E, H
E : B, F
F : A
D : F
H : A

GRAPH 2[Start node =A, Goal node =E]



Adjacency Lists

A : B, D
B : C, F
C : E, G
G : E
E : B, F
F : A
D : F

2. MAZE PROBLEM

S	A	B	C	D	F
#	G	#	H	#	I
#	J	#	#	K	L
#	M	N	#	O	#
#	#	P	Q	R	E

Consider the given state-space graph with start state S and goal state E. Perform BFS and DFS using OPEN and CLOSE lists. Assume that neighboring nodes are inserted into OPEN in the order given in the graph, repeated nodes are not added to OPEN, and the goal test is performed when a node is removed from OPEN. For BFS, use OPEN as a FIFO queue; for DFS, use OPEN as a LIFO stack. Show the contents of OPEN and CLOSE at each step until the goal E is reached.

KEY POINTS :

- **S represents the start**
- **E represents the goal state**
- **# Represents the wall**

1. **First create a search tree**
2. **Apply BFS and DFS**
3. **The solution should contain the order of the traversed nodes, path and path cost.[Assume each direct edge to have cost=1].**